Basics of FIM: Statistical Learning Sample Questions

October 2026

Block 2: Statistical Learning

Task 1: Multiple Choice

Cross the correct answer. In case that you want to change your answer, color the box of the previous answer completely and cross the new correct answer.

- ■ Answer a).
- □ Answer b).
- ☒ Answer c).
- □ Answer d).

There are no penalty points.
Exactly one answer is correct.
For every correct answer you receive **1 point**.

1. In a statistical learning context, let $\hat{f} : X \to Y$ denote the estimated mapping from predictors to response. Which of the following statements describes the typical behavior as the flexibility of $\hat{f}$ increases?

   ☒ Its variance increases and its bias decreases.

   ☐ Its variance decreases and its bias increases.

   ☐ Its total error increases.

   ☐ Its total error decreases.

   Solution:
   More flexible methods follow the training data more closely (higher variance) and impose fewer assumptions on the shape of f (lower bias). The total error is U-shaped in flexibility, so neither of the last two statements holds in general.

2. In a simple linear regression, given an explained sum of squares ESS of 8 and a residual sum of squares RSS of 12, what is the coefficient of determination R^2?

   ☐ 0.75.

   ☐ 0.60.

   ☐ 0.67.

   ☒ 0.40.

   Solution:
   $R^2 = \frac{\text{ESS}}{\text{TSS}}$ with $\text{TSS} = \text{ESS} + \text{RSS} = 20$, hence $R^2 = 8/20 = 0.40$.

3. Let two features x^1 and x^2 be **dependent**, with realizations x^1_m ($m = 1, \ldots, k$) and x^2_s ($s = 1, \ldots, r$). Which of the following expresses their joint relative distribution correctly?

   ☐ $f(x^1_m, x^2_s) = f(x^1_m) + f(x^2_s | x^1 = x^1_m)$.

   ☒ $f(x^1_m, x^2_s) = f(x^1_m) \times f(x^2_s | x^1 = x^1_m)$.

   ☐ $f(x^1_m, x^2_s) = f(x^1_m) \times f(x^2_s)$.

   ☐ $f(x^1_m, x^2_s) = f(x^2_s | x^1 = x^1_m) / f(x^1_m)$.

   Solution:
   For dependent variables, the joint distribution equals the marginal of one variable multiplied by the conditional of the other given it. The product of the two marginals only holds under independence.

Task 2: Python Programming

You fit a simple linear regression $y = \beta_0 + \beta_1 x$ by least squares to three observations. Below is a code snippet with missing parts.

```
1  xd = [1, 2, 3]
2  yd = [2, 4, 5]
3
4  xbar = sum(xd) / len(xd)
5  ybar = sum(yd) / len(yd)
6  sxy = sum((xi - xbar) * (yi - ybar) for xi, yi in zip(xd, yd)
       )
7  sxx = sum((xi - xbar) ** 2 for xi in xd)
8
9  b1_h = _______________ / _______________
10
11 b0_h = _______________ - _______________ * _______________
```

(a) Fill in the missing code in lines 9 and 11 such that `b1_h` and `b0_h` contain the least-squares estimates $\hat{\beta}_1$ and $\hat{\beta}_0$.
Solution:
b1_h = sxy / sxx
b0_h = ybar - b1_h * xbar

(b) State the two values that the `print` statement outputs.
Solution:
slope b1 = 1.50, intercept b0 = 0.67, since $\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} = \frac{3}{2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{11}{3} - 3$.

Task 3: Python Programming

The lists `yt` and `yh` hold the observed responses and the fitted values of a linear regression model. The code below computes three sums of squares but uses uninformative variable names.

```
yt = [1.0, 3.0, 2.0, 4.0]
yh = [1.3, 2.1, 2.9, 3.7]

ybar = sum(yt) / len(yt)

A = sum((yi - ybar) ** 2 for yi in yt)
B = sum((yi - yhi) ** 2 for yi, yhi in zip(yt, yh))
C = sum((yhi - ybar) ** 2 for yhi in yh)

print(f'A = {A:.2f}, B = {B:.2f}, C = {C:.2f}, B + C = {B + C
    :.2f}')
```

(a) Assign each of `A`, `B` and `C` to one of the quantities TSS, RSS and ESS.
Solution:
A = TSS (total sum of squares), B = RSS (residual sum of squares), C = ESS (explained sum of squares). The output is A = 5.00, B = 1.80, C = 3.20, B + C = 5.00, which illustrates TSS = RSS + ESS for a least-squares fit with intercept.

(b) Spell out the abbreviation ESS and describe in one sentence what it quantifies about a linear regression model.
Solution:
ESS stands for explained sum of squares. It measures how much of the total variation of y around its mean is captured by the fitted values, i.e. the share of the variance that the model explains through x.

(c) State the formula for the coefficient of determination R^2 in terms of the quantities `A`, `B` and `C` from the code above.
Solution:
$R^2 = 1 - \frac{B}{A}$, equivalently $R^2 = \frac{C}{A}$ since $A = B + C$. With the values above, $R^2 = 1 - \frac{1.8}{5.0} = 0.64$.